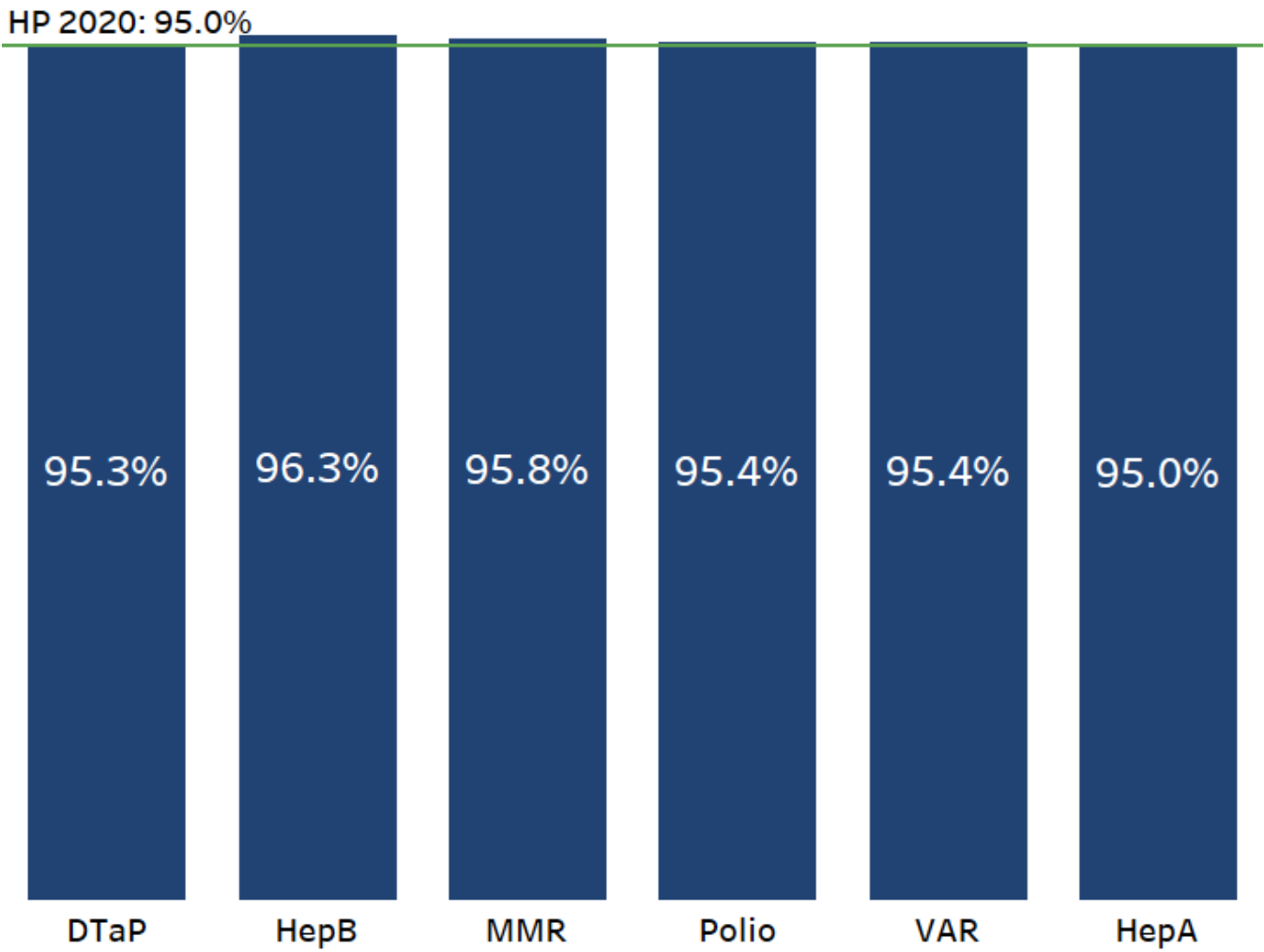


Figure 1. Percentage of public and private school kindergarten students who were complete for each vaccine series, compared to Healthy People 2020 (HP 2020) goal of 95% complete



Public Schools Statewide Summary

Of the 1,011 public schools that were asked to complete the 2021-2022 Kindergarten Assessment 986 (97%) submitted results. Survey participation remains high, yet these figures represent a 2.5% decline in the number of public-school kindergarten programs represented in this report when compared to the 2020-2021 Kindergarten Immunization Compliance Assessment. Statewide, 93.9% of public-school students were fully immunized at the time of the survey (70,274 of 74,830 students). Students who were not fully immunized fell into the following categories, in decreasing order of prevalence (Note: arrows and corresponding figures represent the percent change between 2020-2021 and 2021-2022 assessments):

- Religious Exemption: 2.2% (n=1,653) ↑ 0.5%
- Temporary Certificate: 2.0% (n=1,501) ↑ 1.0%
- Incomplete Record: 1.2% (n=907)
- Missing Record: 0.5% (n=349)
- Medical Exemption: 0.1% (n=82)
- Transfer <30 days: 0.1% (n=85)

Private Schools Statewide Summary

Although our response rate remains relatively stable among private schools, we continue to see a shift towards private schools with an all-time high of 5.4% of all kindergarten enrollment reported in the 2021-2022 Kindergarten Immunization Compliance Assessment. A total of 175 private schools (out of 272 schools; 64%) with kindergarten enrollment completed the 2021-2022 Kindergarten Assessment. Statewide, the survey response rate for private schools increased by 3% resulting in an 11.5% increase in the number of private school kindergarten students represented in this report. Of those students, only 89.4% (3,834 of 4,290 students) were fully immunized, which represents a 2.9% decrease in the fully immunized rate when compared to last year. Unlike the public-school Kindergarten Assessment, the private school survey does not ascertain how many private school students had immunization certificates that were validated by TennHIS. Students who were not fully immunized fell into the following categories, in decreasing order of prevalence (Note: arrows and corresponding figures represent the percent change between 2020-2021 and 2021-2022 assessments):

- Religious Exemption: 4.2% (n=181) ↑ 0.2%
- Incomplete Record: 3.2% (n=139) ↑ 0.9%
- Missing Record: 2.5% (n=107) ↑ 1.5%
- Temporary Certificate: 0.4% (n=18) ↓ 0.3%
- Medical Exemption: 0.3% (n=11) ↑ 0.2%
- Transfer <30 days: N/A for private schools based on data collection methods

Trends over Time

The 10% decline in overall enrollment observed in the 2020-2021 Kindergarten survey was almost completely reversed in the 2021-2022 survey which saw enrollment grow by 7.2% to 79,120 kindergarten students. As mentioned, approximately 5.4% of the kindergarten assessment enrollment in this report are from private schools compared to 3.5% in 2018-2019. Since 2015, the percentage of public and private school students who received all vaccines per Tennessee's state requirements for kindergarten entry (i.e., were "fully immunized") has consistently been close to 95%. In recent years that stability ended with an all-time low of 93.7% of all 2021-2022 reported kindergarten students in Tennessee being fully immunized. For a community to remain at low-risk for a highly infectious vaccine-preventable disease outbreak such as measles, this rate must remain above 95%.⁷ With fewer kindergarteners in the fully immunized category statewide, many counties fail to meet the minimum threshold for vaccine-preventable disease immunity. In fact, the number of counties that meet the minimum 95% fully immunized threshold has decreased from 72 to 43 counties in just two years representing a decrease of 40.3%.

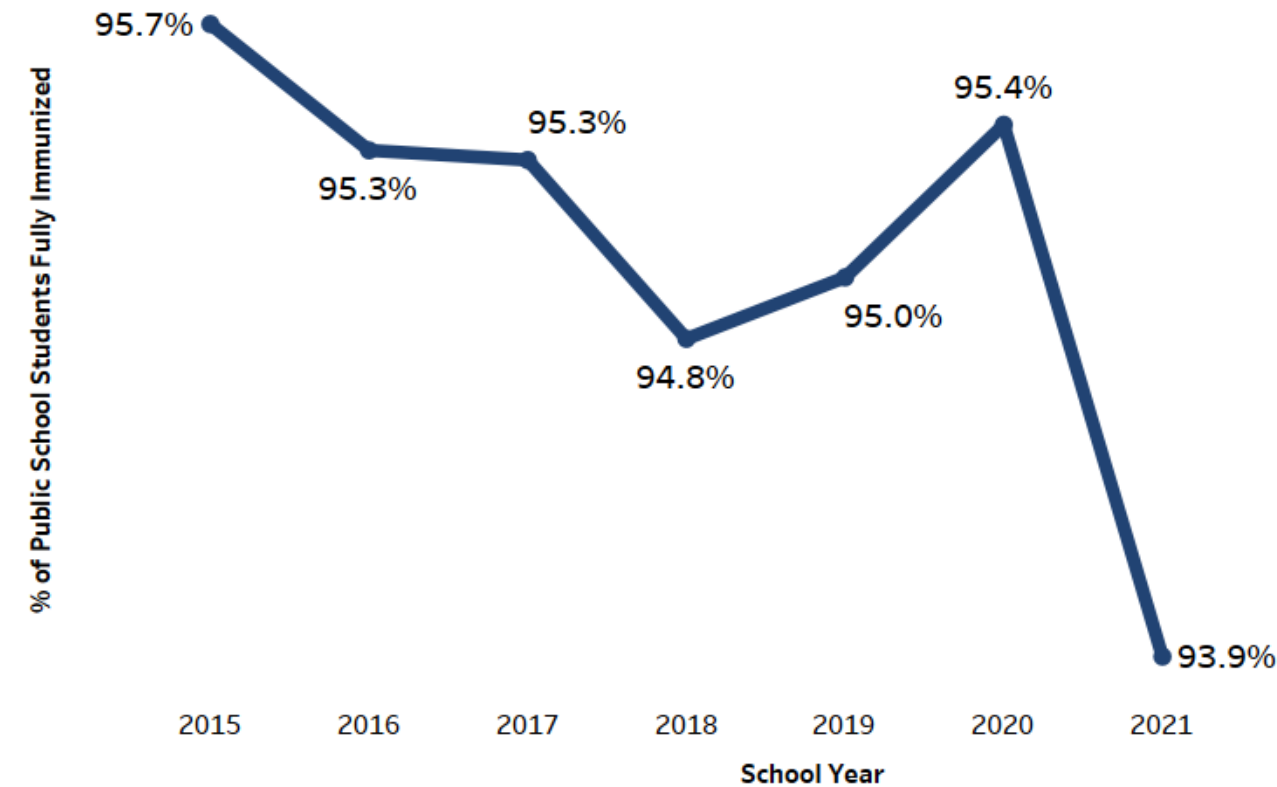
Despite a relatively stable number of exemptions over the past five years, this year (2021-2022), we recorded a 1.4% increase in overall exemptions, statewide. Dating back to 2016 we recorded declines in incomplete records and increases in temporary certificates (see Figure 3). In 2020-2021, this four-year trend reversed with temporary certificates decreasing by 0.9% and incomplete records increasing by 0.5%. This year (2021-2022) temporary certificates rebounded back to the 2019 level of 1.9%, but incomplete records maintained the previous year's increase of 1.3%. Together, the shifts in incomplete records and temporary certificates equate to a higher number of children missing required immunizations for kindergarten enrollment. Furthermore, in 2021-2022 we recorded a 0.5% increase in religious exemptions to an all-time high of 2.3% of all kindergarten students, statewide. The increasing number of kindergarteners who are not fully immunized indicates Tennessee schools and communities are more susceptible to vaccine-preventable disease outbreaks.

⁸ <https://globalhealthnow.org/2019-12/myth-about-herd-immunity>

Impact of COVID Pandemic

National estimates have shown a decline in overall vaccination rates following the United States declaration of a national state of emergency due to the SARS-CoV-2 pandemic. Although sporadic increases in vaccination administration have been observed in the last year, the improvements were neither large nor long-lasting enough to overcome the decline documented during the height of the pandemic. The Kindergarten Immunization Compliance Assessment Report affords a unique opportunity to document the impact of the pandemic upon childhood immunization rates in Tennessee. As pandemic restrictions have been lifted overall kindergarten enrollment increased in 2021-2022 but still fell short of pre-pandemic 2019-2020 figures. Since the pandemic started, the number of counties with public schools that fall below the 95% fully immunized threshold increased 31%. As a result, only 45% (43 of 95) of in the 2021-2022 Kindergarten assessment meet the minimum threshold of 95% fully immunized for vaccine-preventable disease immunity. Of the 31 counties with private school responses to the 2021-2022 survey, only 26% exceed 95% of students fully immunized. While overall immunization coverage has declined, the survey results show religious exemptions in public schools have risen by 0.4% while private school religious exemptions have increased by 1.5% since the 2019-2020 school year. Since the onset of the pandemic, we have observed a sharp decline in fully immunized kindergarten students in Tennessee which represents an increased risk of vaccine-preventable disease outbreaks. Immediate action to increase compliance with recommended childhood immunizations is necessary to decrease the risk of future vaccine-preventable disease outbreaks in our schools and communities alike.

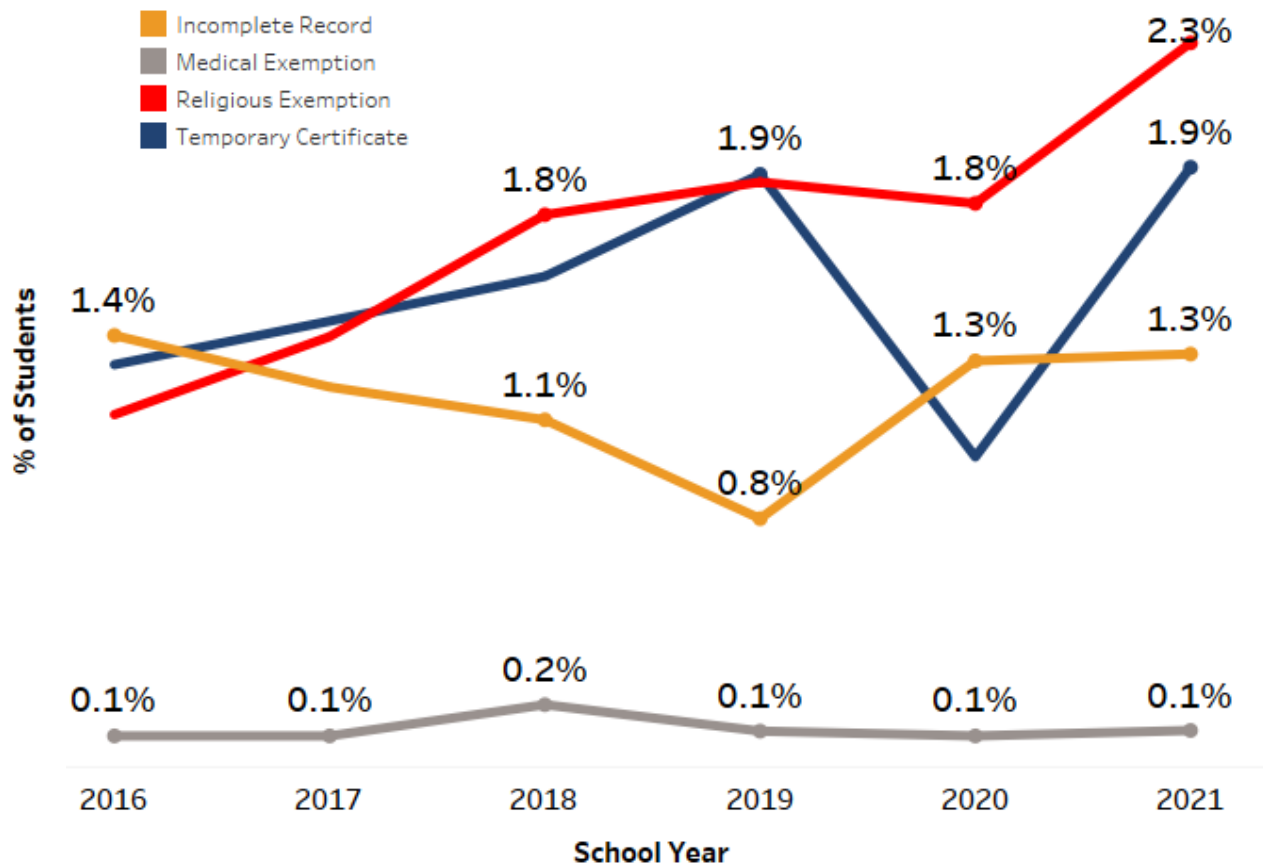
Figure 2. Percentage of public-school students fully immunized, 2015-2021



The percentage of all TN kindergarten students who were not fully immunized, by category, over time, are shown in **Figure 3**. Incomplete immunization records decreased for four years until 2020-2021 when they increased by 0.5% to 1.3% where it remained for the 2021-2022 survey (see Figure 3, gold line). The SARS-CoV-2

pandemic resulted in barriers to obtaining immunizations as well as administrative challenges in the schools responsible for enforcing the state's immunization requirements. Although many of these disruptions have subsided, the subsequent increase in incomplete records remains. Similarly, the trend in temporary certificates (see Figure 3, blue line) suddenly reversed course in 2020-2021, but has returned sharply to pre-pandemic status. Although the driving force behind last year's sudden shift in temporary certificates to incomplete records is unknown, it appears that these indicators (temporary certificates and incomplete records) of unvaccinated kindergartners remain clear contributors to the low fully immunized status of student populations. The percentage of students claiming a religious exemption also increased drastically from 1.8% to 2.3% in 2021-2022 (see Figure 3, red line). The percentage of kindergarten students claiming medical exemptions from a vaccine remained stable at approximately 0.1% each year, consistent with the national rate of medical exemptions (see Figure 3, grey line). The percentage of students with missing vaccine records has also remained relatively stable at approximately 0.6% (not shown in figure). The large increases in temporary certificates and religious exemptions represents the largest contributors to a record high 6.1% of unvaccinated kindergarten students and high vulnerability for vaccine preventable disease outbreaks.

Figure 3. Percentage of all TN kindergarten students not fully immunized, by category, 2016-2021



High immunization coverage rates are critical to avoiding vaccine-preventable disease (VPD) outbreaks (e.g., measles, pertussis). A VPD outbreak may occur when community members are inadequately immunized against a disease. To remain at low risk for a VPD outbreak the immunization rate should remain at or above 95%. In the maps above and tables below, an immunization rate of at least 95% meets this immunization goal, an immunization rate of 90% to 94.9% needs improvement, and an immunization rate below 90% places the community at increased risk for vaccine-preventable disease outbreaks.

For detailed Kindergarten Assessment results at the county, school districts, and school levels, please see the *Public-School Districts (SD) by County Summary* section below.

Table 1. Percentage of **public and private school** students fully immunized, total # students, and the number of schools with students fully immunized within each color category, by county, in descending order by “% students fully immunized” (Color Key: <90%; 90-94.9%; 95-100%)

County	% Students Fully Immunized in County	Total # Students in County	# Schools with <90% Students in County Fully Immunized	# Schools with 90-94% Students in County Fully Immunized	# Schools with 95-100% Students in County Fully Immunized
BENTON	100.0%	172	0	0	3
HANCOCK	100.0%	88	0	0	1
JEFFERSON	99.2%	532	0	0	8
PERRY	98.5%	67	0	0	2
UNICOI	97.9%	144	1	0	3
TIPTON	97.8%	784	0	0	8
DYER	97.7%	472	1	0	5
GRAINGER	97.7%	218	0	1	3
MARION	97.5%	319	0	0	5
SCOTT	97.5%	317	0	0	6
HICKMAN	97.2%	213	0	0	2
JOHNSON	97.1%	138	0	2	2
CROCKETT	97.0%	199	0	1	4
MARSHALL	97.0%	395	0	1	2
MCMINN	97.0%	266	0	0	4
COCKE	96.9%	381	1	2	7
HARDEMAN	96.9%	261	0	1	5
WEAKLEY	96.9%	322	0	1	4

County	% Students Fully Immunized in County	Total # Students in County	# Schools with <90% Students in County Fully Immunized	# Schools with 90-94% Students in County Fully Immunized	# Schools with 95-100% Students in County Fully Immunized
CLAY	96.8%	95	0	0	2
WAYNE	96.8%	154	0	1	2
OVERTON	96.7%	241	0	1	4
LAUDERDALE	96.5%	254	0	1	1
ROANE	96.4%	475	0	1	5
WARREN	96.3%	482	1	1	8
ANDERSON	96.2%	894	3	3	10
CARROLL	96.2%	316	0	1	4
SMITH	96.2%	262	0	1	4
FAYETTE	96.0%	329	0	3	3
LAKE	95.9%	49	0	0	1
DICKSON	95.8%	573	1	2	6
OBION	95.8%	382	0	2	4
STEWART	95.7%	141	0	0	2
WASHINGTON	95.7%	1180	0	8	12
COFFEE	95.6%	726	2	3	7
FRANKLIN	95.6%	428	1	1	7
HOUSTON	95.5%	89	0	0	2
MCNAIRY	95.3%	322	0	1	4
SULLIVAN	95.3%	1462	1	9	14
GREENE	95.1%	697	0	5	6
HAWKINS	95.0%	537	1	4	5
BLEDSON	94.9%	118	0	2	1

County	% Students Fully Immunized in County	Total # Students in County	# Schools with <90% Students in County Fully Immunized	# Schools with 90-94% Students in County Fully Immunized	# Schools with 95-100% Students in County Fully Immunized
CHEATHAM	94.9%	475	1	2	4
WILSON	94.8%	1884	2	9	11
DECATUR	94.7%	113	0	1	1
DEKALB	94.7%	225	0	1	1
HUMPHREYS	94.7%	208	1	0	3
MEIGS	94.7%	150	0	1	1
CANNON	94.6%	149	1	2	3
MACON	94.6%	353	0	1	2
RUTHERFORD	94.5%	4457	4	18	18
TROUSDALE	94.5%	127	0	1	0
WHITE	94.5%	308	1	2	3
LINCOLN	94.4%	356	1	1	4
CARTER	94.3%	507	1	3	7
CLAIBORNE	94.3%	333	1	4	3
GILES	94.3%	317	0	4	0
HARDIN	94.2%	328	1	2	4
UNION	94.2%	241	1	1	3
GIBSON	94.0%	665	2	3	5
HAMBLÉN	94.0%	821	2	5	6
HENRY	94.0%	315	1	2	1
MONTGOMERY	93.9%	3155	2	15	13
RHEA	93.9%	379	1	1	3
ROBERTSON	93.9%	872	3	4	6
LEWIS	93.8%	146	0	1	0

County	% Students Fully Immunized in County	Total # Students in County	# Schools with <90% Students in County Fully Immunized	# Schools with 90-94% Students in County Fully Immunized	# Schools with 95-100% Students in County Fully Immunized
MADISON	93.8%	1145	1	6	8
SUMNER	93.8%	2242	6	9	16
MOORE	93.7%	63	0	1	0
SHELBY	93.7%	12297	34	42	108
LAWRENCE	93.5%	585	1	5	4
HAYWOOD	93.3%	180	0	1	0
HENDERSON	93.3%	403	2	2	4
KNOX	93.1%	5044	16	23	37
SEQUATCHIE	93.1%	160	0	1	0
JACKSON	92.9%	127	0	2	0
DAVIDSON	92.8%	7869	29	34	53
BEDFORD	92.7%	726	3	3	3
PICKETT	92.6%	54	0	1	0
CAMPBELL	92.5%	424	4	1	4
PUTNAM	92.5%	863	4	4	5
BLOUNT	92.3%	1292	5	9	5
WILLIAMSON	92.3%	3216	9	21	10
MORGAN	91.8%	232	2	1	2
MONROE	91.6%	501	2	1	3
SEVIER	91.5%	1210	2	7	5
LOUDON	91.3%	561	2	2	4
BRADLEY	90.9%	1085	9	2	6
CUMBERLAND	90.9%	514	3	4	2
POLK	90.8%	142	1	2	0

County	% Students Fully Immunized in County	Total # Students in County	# Schools with <90% Students in County Fully Immunized	# Schools with 90-94% Students in County Fully Immunized	# Schools with 95-100% Students in County Fully Immunized
HAMILTON	89.7%	3967	27	19	14
CHESTER	89.4%	246	2	0	1
MAURY	88.9%	1172	7	7	3
GRUNDY	88.8%	134	2	2	1
VAN BUREN	82.0%	50	1	0	0
FENTRESS	74.4%	238	4	0	0

Public and Private School Summaries by County and School District

The following series of tables and graphs capture information at two different levels: 1) at the county-level, featuring information for both public and private schools within the county; and 2) at the school district level, featuring information for all schools within the school district. Data are presented by county in alphabetical order, with school districts nested under each respective county in alphabetical order, and schools nested under each respective school district in alphabetical order.

Below are descriptions of each vaccine type shown in the county tables and graphs.

- Diphtheria-Tetanus-Pertussis vaccine (DTaP)⁸
- Hepatitis A Virus vaccine (HAV)⁹
- Poliomyelitis vaccine (IPV or OPV)
- Measles, Mumps, Rubella vaccine (MMR)
- Hepatitis B Virus vaccine (HBV)
- Varicella (chickenpox) vaccine (or credible history of disease)

Below are descriptions of each category shown in the county tables and graphs. All table percentages on a given row are the percentage of the “Total Students” number on that row.

County	Tennessee county name. School districts are grouped into their corresponding county.
School District	Department of Education school district name. Schools are grouped into their corresponding school district. Private schools are listed under a “private” district name within each county where applicable.
School Name	School name, listed alphabetically within each school district.
Total Students	Total number of enrolled kindergarten students at the time of immunization record assessment.
Fully Immunized	Number and percentage who meet state school entry requirements by being fully immunized.
Religious Exemption	Number and percentage with a valid religious exemption.
Medical Exemption	Number and percentage with a valid medical exemption to one or more vaccines.
Temporary Certificate	Number and percentage with a valid temporary certificate.
Excluded Transfer	Students exempt from assessment because records are unavailable due to a recent transfer from another TN school (within 30 days of the assessment). N/A for private schools based on data collection methods.
Incomplete Record	Number and percentage who do not meet requirements because they have an incomplete vaccination record.
Missing Record	Number and percentage who do not meet requirements because the school has no official immunization certificate on record.

⁸ Diphtheria-Tetanus toxoid vaccine (DT), if medically indicated

⁹ <https://www.tn.gov/health/cedep/immunization-program/ip/immunization-requirements/childcare-12th-grade-immunization-requirements.html>